

MAIE 5532 — Sample Midterm Exam 1 (60 minutes)

Weeks 1–5: Frameworks, AD, Graph Optimization, Parallelism, Distributed/Federated Learning

Instructions: Tick the box to the left of each chosen option. Some questions are multi-select.

Name: Type your name here

Student ID: Type your student ID here

Questions (40)

Question 1

In which scenario is forward-mode automatic differentiation the best fit?

- ☐ A. Computing sensitivities w.r.t. a few inputs for a function with many outputs
- ☐ B. Training a deep neural network with millions of parameters and a scalar loss
- ☐ C. Efficiently computing many VJPs (vector–Jacobian products)
- ☐ D. Finite differences for large parameter vectors

Knowledge point: Forward-mode AD is suited to few inputs and many outputs.

Slides: [Week 2 slides, pp. 29–30](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 2

In JAX, which transformation most directly vectorizes a function over a batch dimension without manual loops?

- ☐ A. pmap
- ☐ B. vmap
- ☐ C. jit
- ☐ D. grad

Knowledge point: JAX vectorization with vmap.

Slides: [Week 1 slides, pp. 34–35](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 3

On modern NVIDIA GPUs, which tensor layout often yields more coalesced memory access for many 2D conv workloads?

- ☐ A. NLC
- ☐ B. CHWN

- ☐ C. NHWC
- ☐ D. NCHW

Knowledge point: GPU-friendly tensor layouts and coalesced access (NHWC).

Slides: [Week 3 slides, p. 25](#); [Week 3 slides, pp. 41–42](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 4

Regarding gradient checkpointing in reverse-mode AD, which is most accurate?

- ☐ A. Cuts compute $\sim 50\%$ but increases memory
- ☐ B. Reduces memory $O(\log n)$ at $\sim 10\times$ compute
- ☐ C. Replaces backward with numerical differentiation
- ☐ D. Reduces peak memory ($\approx \sqrt{n}$ layers) with modest compute ($\sim 33\%$)

Knowledge point: Reverse-mode gradient checkpointing and its memory/compute trade-off.

Slides: [Week 2 slides, p. 40](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 5

For a ring all-reduce with payload size P and N devices, the per-device traffic per step (idealized) is:

- ☐ A. $P \cdot (N-1)/N$
- ☐ B. $2 \cdot P/N$
- ☐ C. $2 \cdot P \cdot (N-1)/N$
- ☐ D. $P \cdot (N-1)$

Knowledge point: Ring all-reduce communication volume.

Slides: [Week 5 slides, p. 8](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 6

Which statement about Amdahl's Law is correct?

- ☐ A. With serial fraction $s=0.1$, the maximum speedup is $10\times$ as $N \rightarrow \infty$
- ☐ B. Speedup scales linearly with N for any $s < 1$
- ☐ C. If $s < 1$, the maximum speedup is unbounded
- ☐ D. With $s=0.1$, speedup at $N=10$ must be exactly $10\times$

Knowledge point: Amdahl's Law is not explicitly named in the slides; the closest material covers parallel-scaling limits and sources of scaling loss.

Slides: [Week 4 slides, p. 22](#); [Week 5 slides, p. 14](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 7

Which pipeline scheduler interleaves forward and backward and may introduce weight staleness?

- ☐ A. GPipe (all-forward then all-backward)
- ☐ B. 1F1B
- ☐ C. Synchronous data parallel
- ☐ D. Ring all-reduce

Knowledge point: Pipeline schedules, microbatches, and 1F1B weight staleness.

Slides: [Week 5 slides, pp. 11–12](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 8

Which property defines dual numbers used for forward-mode AD?

- ☐ A. $\epsilon = 0$
- ☐ B. ϵ encodes second-order terms ($\epsilon^2 \neq 0$)
- ☐ C. They implement finite differences
- ☐ D. $\epsilon^2 = 0$ (nilpotency)

Knowledge point: Dual numbers and the nilpotent property $\epsilon^2 = 0$.

Slides: [Week 2 slides, pp. 13–15](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 9

Dual-number multiplication obeys:

- ☐ A. $(a,b) \times (c,d) = (a+c, b+d)$
- ☐ B. $(a,b) \times (c,d) = (ac, bd)$
- ☐ C. $(a,b) \times (c,d) = (ac, ad+bc)$
- ☐ D. $(a,b) \times (c,d) = (a+c, ad-bc)$

Knowledge point: Dual-number multiplication and the product rule.

Slides: [Week 2 slides, p. 8](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 10

Under severe MCU memory limits for on-device learning, which approach is most effective?

- ☐ A. Increase batch size to fill the arena
- ☐ B. Disable backprop in the last k layers
- ☐ C. Store all activations in flash and never recompute
- ☐ D. Aggressive checkpointing with full recomputation during backward

Knowledge point: Checkpointing with recomputation for memory-constrained on-device learning.

Slides: [Week 2 slides, pp. 40–44](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 11**Which collective is bandwidth-optimal for large messages?**

- ☐ A. Ring all-reduce
- ☐ B. Tree all-reduce
- ☐ C. Parameter server pull/push
- ☐ D. All-to-all

Knowledge point: Ring all-reduce bandwidth efficiency for large messages.

Slides: [Week 5 slides, p. 17](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 12**Which framework centers on pure functions transformed by jit/grad/vmap/pmap with immutable arrays?**

- ☐ A. TensorFlow 1.x
- ☐ B. PyTorch (eager + TorchScript)
- ☐ C. TensorFlow 2.x eager + tf.function
- ☐ D. JAX

Knowledge point: JAX functional programming, immutable arrays, and transformations.

Slides: [Week 1 slides, pp. 34–35](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 13**On TPUs, which practice generally improves systolic array utilization for dense layers?**

- ☐ A. Pad/align matrix dims to preferred multiples (e.g., 128)
- ☐ B. Use many tiny GEMMs to vary launch patterns
- ☐ C. Disable XLA layout optimization
- ☐ D. Convert all convs to FFT-based forms by default

Knowledge point: TPU systolic arrays and tile-aligned matrix dimensions.

Slides: [Week 6 slides, p. 12](#); [Week 6 slides, p. 16](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 14

Which permutation converts NHWC → NCHW?

- ☐ A. [0,2,3,1]
- ☐ B. [0,1,3,2]
- ☐ C. [0,3,1,2]
- ☐ D. [3,0,1,2]

Knowledge point: NHWC-to-NCHW tensor dimension permutation.

Slides: [Week 3 slides, p. 25](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 15

Which framework popularized define-by-run dynamic graphs by default?

- ☐ A. TensorFlow 1.x
- ☐ B. PyTorch
- ☐ C. JAX
- ☐ D. Core ML

Knowledge point: PyTorch define-by-run and dynamic computation graphs.

Slides: [Week 1 slides, p. 31](#); [Week 1 slides, p. 41](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 16

In privacy-preserving federated learning:

- ☐ A. DP alone ensures the server sees only encrypted sums
- ☐ B. Secure aggregation hides individuals; DP adds calibrated noise for a formal bound
- ☐ C. Secure aggregation is unnecessary with 8-bit quantization
- ☐ D. DP requires sending raw data to the server

Knowledge point: Secure aggregation and differential privacy have distinct roles in federated learning.

Slides: [Week 5 slides, pp. 47–50](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 17

In elastic distributed training, when world size changes you should:

- ☐ A. Recompute effective global batch and adjust LR accordingly
- ☐ B. Preserve the old process groups indefinitely
- ☐ C. Disable checkpoints to avoid mismatches
- ☐ D. Keep loss scaling fixed regardless of precision

Knowledge point: Elastic distributed training: adapting to world-size changes.

Slides: [Week 5 slides, p. 5](#); [Week 5 slides, pp. 16–20](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 18

Which statement about FSDP (fully sharded data parallel) is correct?

- ☐ A. Keeps full parameters resident on every rank always
- ☐ B. Avoids all-gather in forward
- ☐ C. Only shards optimizer states
- ☐ D. Fully shards params/states; all-gather before compute; reduce-scatter gradients after

Knowledge point: FSDP parameter/state sharding with all-gather and reduce-scatter.

Slides: [Week 5 slides, p. 18](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 19

In TinyML deployments, which is often the dominant energy cost?

- ☐ A. Radio communication (e.g., BLE/Wi-Fi uplink)
- ☐ B. Scalar additions in ALU
- ☐ C. Instruction decoding
- ☐ D. Stack frame setup/teardown

Knowledge point: TinyML energy use and the cost of radio communication.

Slides: [Week 4 slides, p. 35](#); [Week 6 slides, p. 26](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 20

Which stack is hardware-agnostic, imports models, auto-schedules, and can emit standalone C/LLVM for deployment?

- ☐ A. XLA-only
- ☐ B. cuDNN
- ☐ C. TVM
- ☐ D. ONNX as a runtime

Knowledge point: TVM model import, auto-scheduling, and C/LLVM deployment.

Slides: [Week 3 slides, pp. 38–39](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 21

Synchronous data-parallel training with gradient averaging is equivalent to:

- ☐ A. Single device with a smaller batch
- ☐ B. Training only with parameter servers
- ☐ C. Single device only if FP16 is used
- ☐ D. Single device with batch equal to sum of local batches (same optimizer/LR schedule)

Knowledge point: Synchronous data parallelism and global-batch equivalence.

Slides: [Week 5 slides, pp. 6–7](#); [Week 5 slides, p. 16](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 22

In JAX, which transformation compiles a function for optimized execution while preserving semantics?

- ☐ A. grad
- ☐ B. vmap
- ☐ C. jit
- ☐ D. pmap

Knowledge point: JAX jit compilation and XLA optimization.

Slides: [Week 1 slides, p. 34](#); [Week 3 slides, p. 37](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 23

For mixed-precision training stability, which is generally preferred when available?

- ☐ A. FP64 only
- ☐ B. BF16 (wider dynamic range) vs FP16
- ☐ C. INT8 everywhere
- ☐ D. FP16 without loss scaling

Knowledge point: Mixed-precision stability and BF16 dynamic range.

Slides: [Week 6 slides, pp. 10–12](#); [Week 6 slides, p. 22](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 24

Amdahl's Law upper bound on speedup is:

- ☐ A. $1 / (1 - s \cdot N)$
- ☐ B. $(1 - s) / N$
- ☐ C. $1 / (s + (1 - s) / N)$
- ☐ D. $N / (1 + s)$

Knowledge point: The Amdahl speedup bound and serial fraction.

Slides: [Week 4 slides, p. 22](#); [Week 5 slides, p. 14](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 25

Which approach reduces cross-node traffic by keeping heavy reductions inside nodes before inter-node?

- ☐ A. Ring all-reduce
- ☐ B. Tree all-reduce
- ☐ C. Parameter server
- ☐ D. Hierarchical all-reduce

Knowledge point: Hierarchical all-reduce reduces inter-node communication.

Slides: [Week 5 slides, p. 17](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 26

In FedAvg, which weighting reduces bias when aggregating client updates?

- ☐ A. Weight by each client's number of local examples
- ☐ B. Weight all clients equally regardless of data size
- ☐ C. Weight by client battery level
- ☐ D. Weight by client round-trip time

Knowledge point: FedAvg aggregation weighted by client data size.

Slides: [Week 5 slides, p. 6](#); [Week 5 slides, p. 21](#); [Week 5 slides, p. 45](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 27

For small messages, which all-reduce often has lower latency?

- ☐ A. Ring
- ☐ B. Tree
- ☐ C. All-to-all
- ☐ D. Broadcast

Knowledge point: Tree versus ring all-reduce latency for small messages.

Slides: [Week 5 slides, p. 17](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 28

In JAX, which transformation maps a function SPMD-style across devices?

- ☐ A. vmap
- ☐ B. pmap
- ☐ C. grad
- ☐ D. jit

Knowledge point: JAX pmap for SPMD-style multi-device execution.

Slides: [Week 1 slides, pp. 34–35](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answer & explanation](#)

Question 29

(Select all that apply) Effective MCU optimizations for depthwise convolution:

- ☐ A. Precompute index/offset tables to avoid repeated address arithmetic
- ☐ B. Use wider, aligned loads (e.g., 8-byte) when safe
- ☐ C. Loop unrolling to improve instruction-level parallelism
- ☐ D. Frequent malloc/free per row to reduce peak use

Knowledge point: MCU depthwise-convolution kernel optimizations.

Slides: [Week 3 slides, pp. 54–56](#); [Week 6 slides, p. 23](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 30

(Select all that apply) Federated learning client-scheduling practices that reduce round time and bias:

- ☐ A. Over-invite and accept the first N updates before deadline
- ☐ B. Stratified sampling across device or data strata
- ☐ C. Cap per-client participation frequency
- ☐ D. Always wait for all stragglers beyond the deadline

Knowledge point: Federated client scheduling, stratified sampling, and participation fairness.

Slides: [Week 5 slides, p. 22](#); [Week 5 slides, pp. 52–53](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 31

(Select all that apply) Differential privacy in federated learning:

- ☐ A. Client-side norm clipping + adding Gaussian noise before secure aggregation
- ☐ B. Server decrypts individual updates and adds noise later
- ☐ C. Track ϵ , δ with a privacy accountant and stop/adjust at budget limit
- ☐ D. Transport encryption alone suffices for DP guarantees

Knowledge point: Client-level DP: clipping, Gaussian noise, and privacy accounting.

Slides: [Week 5 slides, p. 49](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 32

(Select all that apply) Energy-efficiency strategies for TinyML deployments:

- ☐ A. Duty cycling (short active bursts, long sleeps)
- ☐ B. Cascaded detection (cheap stage gates a heavier classifier)
- ☐ C. Compute more locally to transmit fewer bytes
- ☐ D. Trigger training during active user interaction windows only

Knowledge point: TinyML duty cycling, cascaded inference, and local processing.

Slides: [Week 4 slides, pp. 39–41](#); [Week 6 slides, p. 26](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answers & explanation](#)

Question 33

(Select all that apply) Memory management best practices for TFLM on MCUs:

- ☐ A. Use a fixed tensor arena; avoid heap allocation in the hot path
- ☐ B. Immediate error if arena is too small
- ☐ C. Preallocate and reuse buffers; avoid fragmentation
- ☐ D. Prefer malloc/free per inference for flexibility

Knowledge point: TFLM fixed tensor arenas, fail-fast behavior, and buffer reuse.

Slides: [Week 1 slides, p. 55](#); [Week 3 slides, p. 43](#); [Week 6 slides, p. 23](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answers & explanation](#)

Question 34

(Select all that apply) Pipeline parallelism statements:

- ☐ A. Bubble fraction per step $\approx (m-1)/b$
- ☐ B. 1F1B reduces bubbles/activation footprint but may introduce weight staleness
- ☐ C. GPipe interleaves forward/backward by design
- ☐ D. Increasing microbatches reduces bubbles but can increase memory/latency

Knowledge point: Pipeline bubbles and the GPipe/1F1B schedule trade-offs.

Slides: [Week 5 slides, pp. 11–12](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 35

(Select all that apply) Phases typically present in hierarchical all-reduce:

- ☐ A. Intra-node reduce-scatter
- ☐ B. Inter-node reduce-scatter
- ☐ C. Inter-node all-gather
- ☐ D. Intra-node all-gather

Knowledge point: Hierarchical all-reduce phases: intra/inter-node reduce-scatter and all-gather.

Slides: [Week 5 slides, p. 17](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 36

(Select all that apply) ZeRO optimizer stages:

- ☐ A. Stage 1: shard optimizer states
- ☐ B. Stage 2: shard optimizer states + gradients
- ☐ C. Stage 3: shard parameters as well (optionally offload)
- ☐ D. ZeRO eliminates the need for collectives entirely

Knowledge point: ZeRO stages shard optimizer states, gradients, and parameters.

Slides: [Week 5 slides, p. 18](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 37

(Select all that apply) FSDP behavior:

- ☐ A. All-gather param shards before compute; free promptly; reduce-scatter grads after
- ☐ B. Works with mixed precision to reduce memory
- ☐ C. Requires full params to be present on all ranks at all times
- ☐ D. Wrapping/granularity choices impact comm/compute balance

Knowledge point: FSDP just-in-time all-gather, freeing parameters, and gradient reduce-scatter.

Slides: [Week 5 slides, p. 18](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Question 38

(Select all that apply) Reproducibility at scale:

- ☐ A. Seed global and per-device RNGs (Python/NumPy/framework/CUDA)
- ☐ B. Prefer nondeterministic kernels to improve speed
- ☐ C. Save RNG states and dataloader epoch/offsets in checkpoints
- ☐ D. Pin versions (CUDA/cuDNN/NCCL, drivers, framework) and log topology/world size

Knowledge point: Large-scale reproducibility: RNG seeds, checkpoint state, and environment metadata.

Slides: [Week 5 slides, pp. 19–20](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answers & explanation](#)

Question 39

(Select all that apply) Reducing serial fraction s / hiding comm for better scaling:

- ☐ A. Overlap communication with compute (bucketization, fused kernels)
- ☐ B. Lower-precision or compressed gradients with error feedback
- ☐ C. Increase local batch judiciously; consider partial/delayed sync variants with care
- ☐ D. Ignore stragglers and disable health checks

Knowledge point: Scaling strategies: overlap communication, compress updates, and tune synchronization.

Slides: [Week 5 slides, p. 7](#); [Week 5 slides, pp. 14–17](#); [Week 5 slides, p. 23](#) **Review PDF:** [All relevant slides, including other answer choices](#)

[Show answers & explanation](#)

Question 40

(Select all that apply) Memory bandwidth optimization in kernels:

- ☐ A. Align loop order with layout to read contiguous memory (NHWC vs NCHW)
- ☐ B. Prefer many tiny kernel launches for better "interleaving"
- ☐ C. Use in-place ops where safe to halve traffic
- ☐ D. Use memory pools/arenas to avoid runtime malloc/free overhead

Knowledge point: Memory-aware kernels: layout-aligned access, in-place operations, and memory pools.

Slides: [Week 3 slides, pp. 25–26](#); [Week 6 slides, pp. 67–69](#) **Review PDF:** [All relevant slides, including other answer choices](#)

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Answers & Explanations

Question 1

A — Forward mode scales with inputs; best for few inputs, many outputs.

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Question 2

B — vmap vectorizes across a batch dimension.

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Question 3

C — NHWC often coalesces better for many conv kernels.

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Question 4

D — Checkpointing reduces peak memory ($\sim\sqrt{n}$) with $\sim 33\%$ compute overhead.

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Question 5

C — Ring traffic per device $\approx 2 \cdot P \cdot (N-1)/N$.

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Question 6

A — Amdahl bound with $s=0.1 \rightarrow \max \sim 10\times$.

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Question 7

B — 1F1B interleaves and can introduce weight staleness.

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Question 8

D — Dual numbers: $\epsilon^2 = 0$ (nilpotent).

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Question 9

C — $(ac, ad+bc)$ implements product rule.

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Question 10

D — Recompute activations during backward to save memory.

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Question 11

A — Ring is bandwidth-optimal for large messages.

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Question 12

D — JAX functional transforms over immutable arrays.

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Question 13

A — Align dims to TPU tiles (e.g., 128) for utilization.

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Question 14

C — NHWC $\rightarrow$ NCHW is [0,3,1,2].

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Question 15

B — PyTorch popularized define-by-run.

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Question 16

B — Secure aggregation hides individuals; DP adds noise & accounting.

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Question 17

A — Adjust LR to new effective global batch; rebuild groups.

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Question 18

D — FSDP shards params/states; all-gather JIT; reduce-scatter grads.

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Question 19

A — Radios often dominate energy cost.

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Question 20

C — TVM imports/auto-schedules/emits C/LLVM.

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Question 21

D — Equivalent to single device with global batch = sum of local batches.

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Question 22

C — jit compiles a function for optimized execution.

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Question 23

B — BF16 preferred for stability vs FP16.

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Question 24

C — Amdahl bound: $1 / (s + (1-s)/N)$.

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Question 25

D — Hierarchical all-reduce: intra-node, then inter-node, then local broadcast.

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Question 26

A — Weight by client sample counts.

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Question 27

B — Tree lower latency for small messages.

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Question 28

B — pmap distributes SPMD across devices.

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Question 29

A, B, C — Precompute indices; aligned wide loads; unroll loops.

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Question 30

A, B, C — Over-invite/accept first N; stratify; cap frequency.

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Question 31

A, C — Client-side clip+noise; privacy accountant for ϵ, δ .

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Question 32

A, B, C — Duty cycle; cascades; compute locally to save radio.

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Question 33

A, B, C — Fixed arena; fail fast if too small; reuse buffers.

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Question 34

A, B, D — Bubble $\approx (m-1)/b$; 1F1B reduces bubbles but may add staleness; more microbatches reduce bubbles with memory/latency trade-offs.

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Question 35

A, B, C, D — Hierarchical reduce-scatter/all-gather intra-/inter-node.

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Question 36

A, B, C — ZeRO stages 1/2/3 shard states, grads, then params (offload optional).

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Question 37

A, B, D — FSDP all-gathers JIT, supports mixed precision, wrapping granularity matters.

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Question 38

A, C, D — Seed RNGs; save RNG/dataloader state; pin versions/topology.

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Question 39

A, B, C — Overlap comm/compute; compress gradients; adjust batch/sync variants carefully.

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Question 40

A, C, D — Contiguous access, in-place where safe, memory pools; avoid many tiny launches.

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